COUNTERS VS ACCUMULATORS

Chapter 7 – Part 2

POTENTIAL ERROR CONDITION

Names to call it

Endless loop

Infinite loop

example

Cout << “enter hours worked: “;

Cin >> hourswrkd;

While (hourswrked > 0)

{

- Hourpay = hourswrked * 35.00;
- Cout << “ your hourly rate is: “ << hourpay;
- } //endwhile

POTENTIAL ERROR CONDITION EXPLAINED

Why?

In the code on the right side, the hourswrked never gets updated. The loop will not end.

example

Cout << “enter hours worked: “;

Cin >> hourswrkd;

While (hourswrked > 0)

{

- Hourpay = hourswrked * 35.00;
- Cout << “ your hourly rate is: “ << hourpay;
- } //endwhile

1. WRITE A C++ WHILE CLAUSE TO:

Process the loop body as long as the value in *quantity* is greater than 0

1. ANSWER

```
while (quantity > 0)
```

2. WRITE A C++ WHILE CLAUSE TO:

Stop the loop when the value in quantity is less than 0

2. ANSWER:

while (quantity < 0)

QUESTION: WHICH OF THE FOLLOWING IS A GOOD SENTINEL VALUE FOR A PROGRAM THAT INPUTS A TEST SCORE?

A. -9
B. 32
C. 45.5
D. 7

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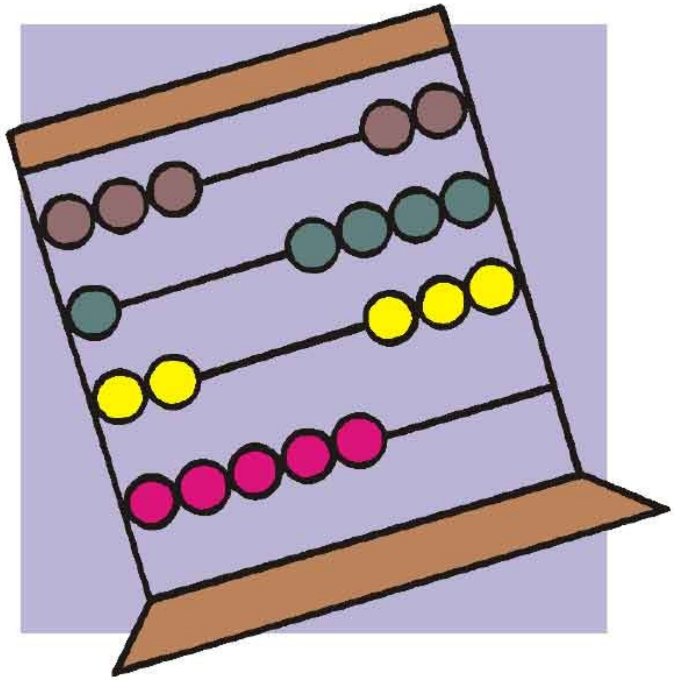
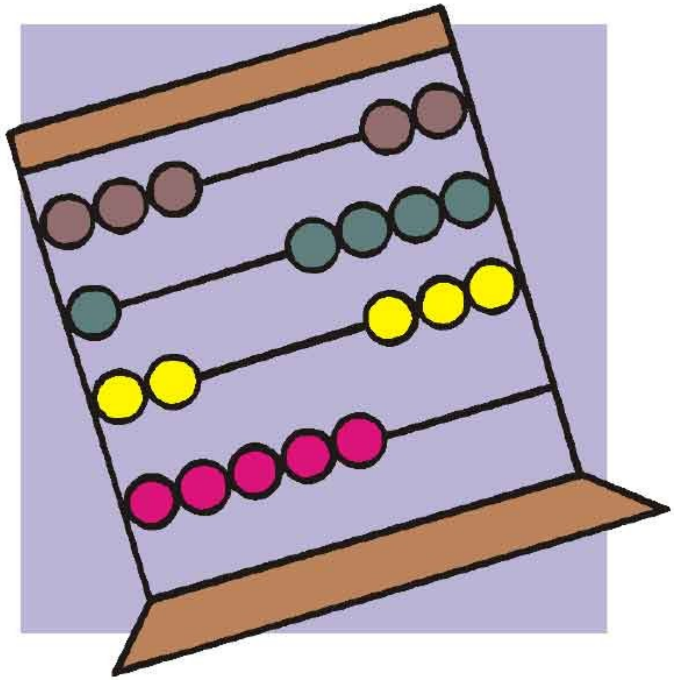
COUNTERS AND ACCUMULATORS



COUNTERS

Are for counting

ACCUMULATORS



Are for accumulating amounts!

COUNTERS AND ACCUMULATORS NEED:

Initializing

- Usually to 0
- But always go by the algorithm
- Only done once

Updating

- Also called incrementing
- Adding a number to the counter or accumulator
- Counters are always updated by a constant number
- Accumulators can be updated by varying amounts
- The updating is done in the body loopy

NOW WE CAN HAVE A COUNTER-CONTROLLED LOOP!

These do not use a sentinel value.

What is a sentinel value?

WHAT IS A SENTINEL VALUE?

A value that ends a loop once it is entered.

EXAMPLE

Int numcnt = 0;

While (numcnt < 10)

{

- Cout >> “The number count is: “ >> numcnt >> endl;
- Numcnt = numcnt +1;
- } //endwhile

YOUR TURN!